

Project Documentation

Project Title: Advanced Bank Data Analytics & Business Intelligence

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Tools Used: Oracle SQL, Power BI Desktop

1. Project Introduction & Problem Statement

The primary objective of this project is to analyze bank customer demographic and financial account data to extract critical Key Performance Indicators (KPIs) that support strategic decision-making for executive bank management.

The raw source data suffered from two main operational challenges:

1. **Data Quality Issues:** The presence of a trailing dot in job categories (e.g., admin.) led to data duplication and inaccurate grouping during aggregation.
2. **Siloed Data Structure:** Essential analysis data was fragmented across multiple disconnected source tables (Demographic accounts table and ATM card activation details table).

2. SQL Data Cleaning & Engineering Phase

The **Oracle SQL** environment was utilized to cleanse, transform, and engineer the dataset directly during the query phase (*On the fly*). This method optimized performance and eliminated the need for allocating additional database storage:

- **String Standardization:** Applied the `RTRIM(JOB, '.')` function to dynamically strip out trailing dots, standardizing all job classifications.
- **Relational Joins:** Executed a structured `INNER JOIN` operation to seamlessly merge the bank database with the customer card details table, mapped securely via the primary relational key (`CUST_ID`).

3. Exploratory Data Analysis & Advanced Analytical Insights

Advanced SQL syntax and **Window Functions** were implemented to calculate complex data trends and segment customer financial behaviors:

- **Segment Ranking:** Utilized `RANK() OVER (PARTITION BY JOB ORDER BY BALANCE DESC)` to identify and isolate the top-depositing clients within every occupational sector.
- **Comparative Aggregations:** Applied the `AVG() OVER` function to benchmark individual customer balances directly against the global bank average. This

exposed high-value customer segments—specifically administrators and retirees—offering clear guidance for targeted banking campaigns.

4. Interactive Power BI Dashboard Design

The optimized Oracle SQL dataset was integrated into **Power BI Desktop** via a direct query connection. Custom **DAX Measures** were written to power an enterprise-grade visual analytics dashboard:

- **Core DAX Measures Built:** Total Customers using explicit COUNT functions, Total Bank Balance utilizing SUM expressions, and Average Customer Balance calculated via AVERAGE syntax.
- **UI/UX & Corporate Design:** Adhered strictly to standard corporate banking aesthetics (elegant navy blue borders framed beautifully against a crisp, low-strain light gray canvas backplate).
- **Visual Hierarchy:** Organized spatial reports symmetrically by displaying an interactive Job Segmentation Column Chart, a distinct Donut Chart for ATM Card distributions, and an operational education filter Slicer.